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1. Title of the Invention

A System and Method for Determining the Length-Measurement Offset of Each Roadside Inductive-Loop Detector in a Chain and Recalibrating Its Vehicle-Length Classification, Using the Invariance of Vehicle Length Between Adjacent Detectors.

2. Field / Area of Invention

The present invention relates to roadside vehicle-detection hardware, in particular inductive-loop detector sites that measure the length of each passing vehicle and transmit counts of vehicles by length class to a traffic-data archive. More particularly, it pertains to a system and method that determine, from those class counts alone, the length-measurement offset in centimetres of each detector in a chain of such sites and the class bias it causes at every class boundary, and correct the recorded counts, using the invariance of vehicle length between adjacent detectors that observe the same vehicles.

3. Prior Patents and Publications from Literature

Table A lists the closest art found, eight publications and six patents, each read in full text before any statement about it was written. Entry 1 is the closest to the invention as a whole; entries 2 to 5 concern how length-classifying loop detectors are calibrated and where their length measurement fails; entries 6 to 8 concern fault screening and the calibration of sensors against their neighbours; entries 9 to 14 are patents on loop detectors, their classification and the quality of their data.

#	Reference	Year / type	Core contribution	Limitation with respect to the present invention
1	Azin and Yang, Multi-Stage Algorithm for Detection-Error Identification and Data Screening, Utah DOT report UT-20.15	2020 / research report	On aggregated detector data, compares the average effective vehicle length (AEVL) distribution of each station with adjacent stations, singles out the station that differs, uses its upstream and downstream neighbours to decide which is faulty, and marks it for maintenance or recalibration; notes that on-ramps alter AEVL downstream. Cites Lu, Yang and Chang (2014, TRR 2443) for the comparison of AEVL distributions.	Closest art; comparison, localisation and flagging are conceded. It does not size a signed error per detector at each class boundary, does not correct any recorded count, uses AEVL, a speed-estimation quantity that mixes speed, occupancy and length, rather than class shares, and does not date a change in a detector's length measurement.
2	Nihan, Wang and Cheevarunothai, Improving Dual-Loop Truck (and Speed) Data, WSDOT WA-RD 647.1	2006 / research report	Identifies dual-loop sensitivity problems from high-resolution per-vehicle loop event data and retunes loop sensitivity, using the short-vehicle length distribution of Wang and Nihan (2004) as ground truth; the ALEDA tool runs on a laptop at the controller cabinet.	Needs per-vehicle event data, an external reference distribution and a visit to each station, one at a time. The invention uses the class counts already archived, with the neighbouring detectors as reference.

#	Reference	Year / type	Core contribution	Limitation with respect to the present invention
3	Minge et al., Loop- and Length-Based Vehicle Classification, FHWA Pooled Fund TPF-5(192), MnDOT report 2012-33	2012 / research report	Recommends probe-vehicle runs as the primary calibration: vehicles of known speed and length are driven through the site, the detector is adjusted to match, and several runs are averaged; documents causes of length error.	Calibration by attendance with known vehicles at each site. No use of neighbouring detectors and no correction of recorded counts.
4	Coifman and Kim, Speed estimation and length based vehicle classification from freeway single-loop detectors, Transportation Research Part C	2009 / journal	Estimates speed and classifies vehicles by length from single-loop on-times, including a recalibration that uses successive vehicles' on-time ratios.	Works within one detector's own per-vehicle data; no comparison between detectors and no class-level correction of an archive.
5	Wu and Coifman, Improved vehicle classification from dual-loop detectors in congested traffic, Transportation Research Part C	2014 / journal	Shows that dual-loop length measurement assumes negligible acceleration and degrades in congestion; bounds the length error from speed and acceleration.	Basis of the free-flow condition used in admission; conceded in full.
6	Lu, Varaiya, Horowitz and Palen, Faulty loop data analysis/correction and loop fault detection, California PATH (manuscript; file dated 2008)	2008 / technical paper	Reviews loop fault detection and data correction, including a test on the ratio of upstream and downstream flows and on-time threshold tests.	Tests on flows and on-times; no length-class comparison, no bias per class boundary, no reclassification.
7	Bychkovskiy, Megerian, Estrin and Potkonjak, A Collaborative Approach to In-Place Sensor Calibration, Proc. IPSN	2003 / conference	Relative calibration of neighbouring sensors that observe the same phenomenon, made consistent over groups; absolute calibration needs a calibrated subset.	General principle of the chain determination, conceded. Continuous sensed field; no class boundaries, no vehicle counts, no reclassification.
8	Balzano and Nowak, Blind Calibration of Sensor Networks, Proc. IPSN	2007 / conference	Recovers unknown sensor gains from routine measurements when the signals are oversampled; the offset component in the signal subspace is generally not identifiable.	Consistent with the need for a reference constant per chain; no traffic detectors, no class boundaries.
9	GB 2385138 A, Validation and calibration of vehicle loop detection systems	2003 / patent	Switched validation loops placed over loop sensors to simulate a vehicle. Its background names loop geometry, turns, detector threshold behaviour and installation irregularities as causes of systematic length over- or under-reading not revealed by inspection.	Hardware test apparatus at the site. Background to the technical problem.
10	EP 2790165 A1, Quality determination in data acquisition	2014 / patent	Withdrawn application. Models presence-pulse sequences with regard to sensor topology and environment; classifies and corrects errors; recommends detector sensitivity settings; trained models including a truck detector.	Pulse-level data and learned models; no comparison of length-class shares of detectors seeing the same vehicles; no bias per class boundary.
11	US 11,461,692 B2, Single-loop vehicle classification using inductive loop vehicle signatures	2022 / patent	Classifies vehicles from single-loop signature waveforms against a signature library with a nearest-neighbour classifier; notes that length-based methods need re-calibration between stations.	Per-site classification from signatures; no neighbour reference; no correction of recorded class counts.
12	US 6,999,886 B2, Vehicle speed estimation using inductive vehicle detection systems	2006 / patent	A single-loop detector card synthesises a second output to mimic a two-loop speed trap so a field controller can estimate speed and length.	Measurement at one site; no cross-site calibration.

#	Reference	Year / type	Core contribution	Limitation with respect to the present invention
13	US 6,337,640 B2, Inductive loop sensor for traffic detection	2002 / patent	Figure-of-eight loop across the lane detects wheels for classification by axle count.	Sensor geometry; classification by axles, not length; no calibration between sites.
14	US 7,764,197 B2, System and synchronization process for inductive loops in a multilane environment	2010 / patent	Schedules the sampling of contiguous loops in adjacent lanes so they do not overlap; signature information includes vehicle length.	Cross-talk between lanes at one site; no comparison of length classes between sites.

Table A: Prior Art Considered

Patent searches were run on Google Patents and FreePatentsOnline with the terms vehicle length, effective vehicle length, length class and classification, combined with adjacent, neighbouring, upstream and downstream detector, and with calibration and correction, in English and Chinese.

4. Summary and Background of the Invention (Gap / Novelty)

Motorway networks classify traffic by vehicle length at roadside inductive-loop sites, and the class counts they archive feed pavement, freight and traffic-management analyses. Each site reads length with an offset peculiar to its installation, so that consecutive sites counting the same vehicles report shares of vehicles below 520 cm differing by a median of 5.0 points on one corridor and 5.8 on another. Current practice either calibrates one site at a time with probe vehicles, test apparatus or an external length distribution, or flags an inconsistent site for later attendance, and the recorded counts stay wrong. The invention uses the vehicles passing consecutive detectors as a moving reference: it solves for each detector's signed class bias at every boundary along a chain of detector pairs, converts those biases into one physical length offset in centimetres through the traffic stream's own length distribution, and reclassifies the recorded counts with every interval total kept. A flag for inspection thereby becomes a corrected record and, for each detector unit, a length offset and a fault class that say what is wrong with the unit and by how much.

4.1 Background and technical problem

A dual-loop detector site computes the speed of a vehicle from the times at which it reaches the two loops of a lane and its length from the time it occupies a loop, less the loop length. The result depends on the stored loop dimensions and spacing, on the path of vehicles over the loops, on the number of turns of each loop, on reinforcement near the loops and on the threshold behaviour of the detector card; entry 9 names each of these as a cause of systematic over- or under-reading of length that inspection does not reveal. A site that reads long moves vehicles from each length class into the next. Its total count stays right, and its fleet mix goes wrong.

Finding the error has meant going to the site. The practice recommended for length-based classification is to drive probe vehicles of known length through the detector and adjust it until it agrees (entry 3); alternatives bring a validation trailer (entry 9) or a laptop that records per-vehicle event data at the controller cabinet and compares it with an external length distribution (entry 2). Each serves one site per visit, on a live carriageway.

The error is large and common. On carriageway A of the M25 in March 2025, 112 pairs of consecutive detectors have no slip road between them and a median ratio of total counts of 1.0010, so they count the same vehicles; their shares of vehicles below 520 cm differ by a median of 5.0 points and by 18.2 at the 90th percentile. Each pair's difference in March correlates with the same pair's difference in June at $r = 0.987$, and hardly at all with the ratio of the two detectors' speeds ($r = -0.07$). On carriageway A of the M1 the figures are 256 pairs, 5.8 and 20.1 points.

The art that does compare neighbouring detectors stops at a flag. Entry 1 singles out the station whose length statistic differs from its neighbours and marks it for recalibration; entries 7 and 8 calibrate generic sensors against one another but concern neither vehicle counts nor class boundaries. None determines how far each detector's classification is wrong at each boundary, none recovers a physical length offset from class counts, and none corrects the record.

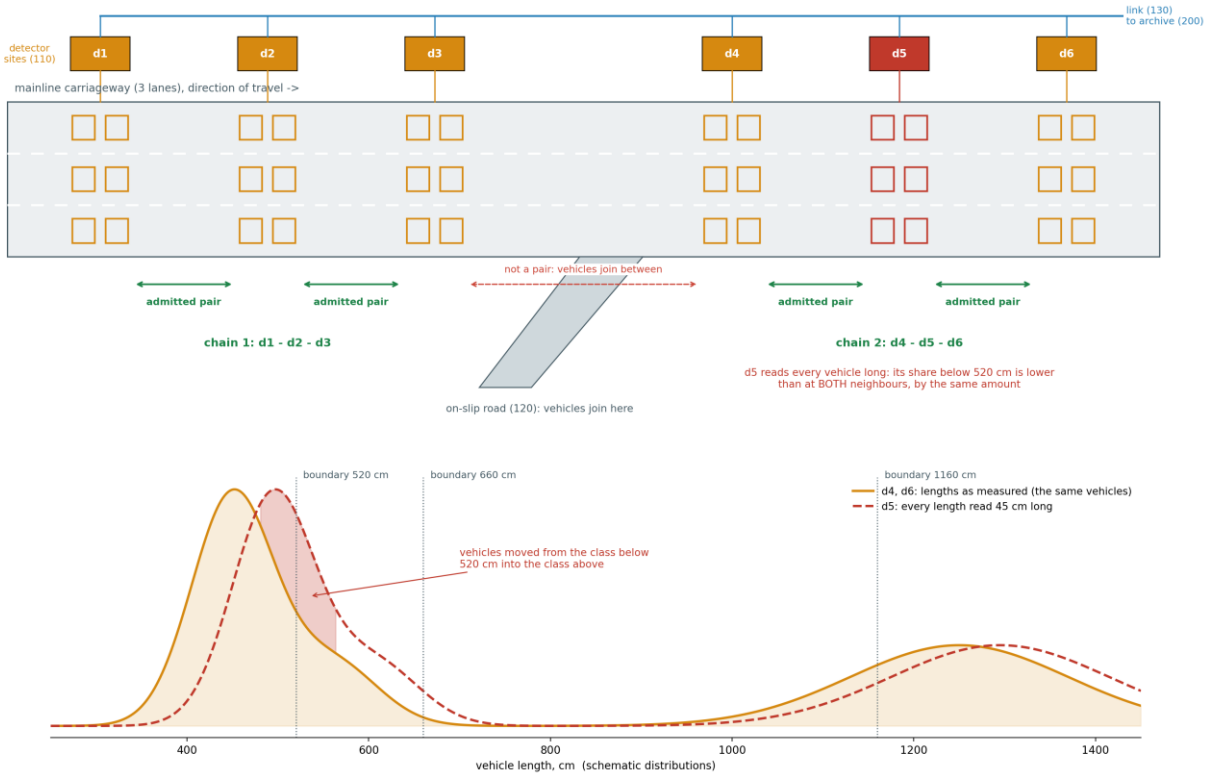


Figure 1. Roadside detector sites on a carriageway and the length-invariance principle. Six detector sites (110), each a loop pair in every lane with a roadside detector card, report class counts over the link (130). Consecutive sites with no slip road between them form admitted pairs and two chains; a site reading every length 45 cm long (d5) moves vehicles across each boundary in proportion to the length density there, and disagrees with both neighbours by the same amount. The length distributions are schematic.

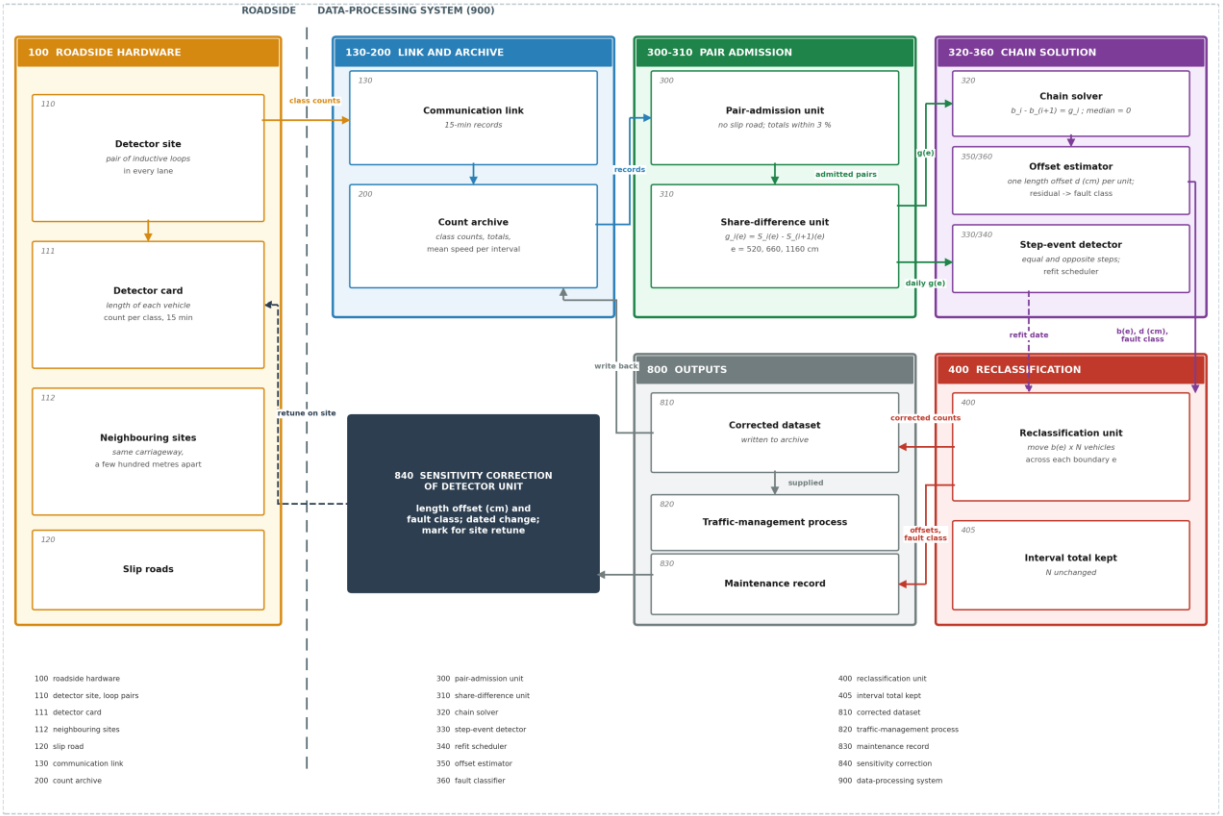


Figure 2. Architecture of the roadside hardware and the data-processing system. Class counts cross from the roadside hardware (100) to the data-processing system (900), pass through pair admission (300, 310), chain solution (320) and the offset estimator (350, 360), and are reclassified (400); outputs (810-830) are the corrected archive, supply to traffic management and the maintenance record, and the sensitivity correction (840) returns to the detector unit when it is retuned.

4.2 Statement of inventive step (non-obviousness)

No single element carries the inventive step. Comparing a length statistic between adjacent detectors and flagging the odd one is entry 1; calibrating sensors against their neighbours is entries 7 and 8; retuning a loop from the length distribution with per-vehicle data is entry 2. The invention resides in the following combination and in effects that no element produces alone:

(a) the invariant: the share of vehicles below each class boundary of the detectors' own classification, compared between two detectors over matched intervals, on the class counts already archived;

(b) admission of a pair only where no slip road lies between the detectors and their totals agree, the physical condition for the two detectors to measure the same vehicles;

(c) determination along a chain of admitted pairs of a signed bias for every detector at every boundary, with the chain's constant fixed by its median detector or by a verified site;

(d) conversion of each detector's biases into one physical length offset in centimetres, through the cumulative length distribution of the traffic stream interpolated from the class shares of the same chain. This rests on a structure found in the data and not taught by any reference: a single offset per detector explains 97.2 % to 98.8 % of the variance of all detectors' biases at all boundaries in every corridor-month, and the ratio of biases between the 660 cm and 520 cm boundaries, 0.15 to 0.19, lies close to the ratio of the stream's length density at those boundaries, predicted from the class shares alone at 0.18 to 0.22. The departure from a single

offset classifies the fault, and the speed bias along the chain separates a loop-spacing error from a detection-threshold error;

(e) correction by reclassifying, in every interval, a number of vehicles equal to the bias times the interval total, which keeps every total; measured better than the correction proportional to the share that the geometry of an offset suggests, and better than a correction from the single offset alone at the 660 and 1160 cm boundaries; and

(f) attribution of a change in length measurement to one detector and one day from equal and opposite steps in its two pairs, with refitting each period and from the day of the change.

The offset of (d) is what a site visit would otherwise measure. It is obtainable only because (a) compares class shares at the boundaries where the length error acts, (b) guarantees that both detectors sample one length distribution, and (c) yields each detector's bias at several boundaries at once; the same chain then supplies the length distribution itself, so no external reference or per-vehicle record is needed. Entry 2 reaches a sensitivity setting only with per-vehicle lengths and a fixed external distribution at one station; entry 1 reaches only a flag. The measured result is a corrected record and a physical parameter for every unit: in months the fit did not see, the mean disagreement between adjacent detectors falls from 7.0-8.9 to 0.58-2.65 points over the pairs touching a corrected detector, and by 57 % to 79 % over all admitted pairs; each unit carries a length offset, with a median of 31 to 36 cm.

5. Objectives of the Invention

1. To provide a roadside detector system that determines each inductive-loop detector's length-measurement offset in centimetres from its archived class counts.

2. To enable correction of each detector's recorded and incoming class counts, keeping every interval total, so that adjacent detectors measuring the same vehicles agree.

3. To provide each physical detector unit with a dated sensitivity correction and fault class, without test vehicles, site apparatus or per-vehicle data.

6. Working Principle (in Brief)

A vehicle's length does not change between two roadside detector sites a few hundred metres apart. Where no slip road joins or leaves the carriageway between them, both sites observe the same vehicles, and the share of those vehicles shorter than any given length is the same at both. A persistent difference between the shares the two sites report can only arise in their measurement.

Each site measures length with a pair of inductive loops in every lane and a detector card that times each vehicle's passage. An installation-specific error adds an offset d to every length the card computes. At a class boundary e, a site that reads long counts below e only the vehicles shorter than e - d, so its share below e falls by the share of vehicles whose length lies between e - d and e; the vehicle count itself is untouched.

Two consecutive sites are paired only if no slip road lies between them and, over the intervals both reported in free-flowing traffic, their total counts agree. Agreement of totals confirms that the pair sees the same vehicles; total counts are used for this admission and for nothing else.

For an admitted pair (i, i+1) and a boundary e, $S_i(e)$ is the share of vehicles that detector i reports below e and $b_i(e)$ the error in that share. The true shares being equal, the measured difference is the difference of the two biases: $g_i(e) = S_i(e) - S_{(i+1)}(e) = b_i(e) - b_{(i+1)}(e)$.

Admitted pairs that share a detector form a chain. A chain of k+1 sites gives k such relations per boundary in k+1 unknowns, so the biases are fixed up to one constant, set by taking the median site of the chain as correct, or a site whose length measurement has been verified. Each interior site belongs to two pairs, and its own error appears in both with opposite signs, which localises it.

Because the vehicles are the same along the chain, one cumulative length distribution F describes them all, and it is known at the class boundaries from the shares the chain reports. A detector with offset d has bias F(e - d) - F(e) at every boundary e: one number per detector, scaled at each boundary by the length density there. Solving this relation at the most populated boundary gives d in centimetres, a physical property of the detector unit.

At the other boundaries the same d predicts the bias. Where the detector's biases depart from that prediction by more than a limit, its fault is not a length offset, for instance a fault confined to one lane or to one class, and the unit is classified accordingly.

A loop-spacing error scales the speed the detector computes and, through it, the length; a detection-threshold (sensitivity) error changes the time a vehicle occupies the loop and so the length alone. The speed bias of each detector, solved along the chain from the ratio of mean speeds in each pair in the same way as the share biases, separates the two causes of an offset.

The recorded counts are corrected by reclassifying, in each interval, a number of vehicles across boundary e equal to b_i(e) times the interval total; the corrected share becomes S_i(e) - b_i(e) and the total count and mean speed are not changed. This per-boundary form was measured better than a correction proportional to the share and better than one computed from the offset alone.

A site whose length measurement changes on a given day, through a retune, a repair or a fault, produces on that day a step of one sign in the pair where it is downstream and an equal step of the other sign in the pair where it is upstream, at every boundary; a change in traffic moves both sites of a pair together and leaves their difference unchanged.

Biases and offsets are refitted each period, and a site with a dated change is refitted from the day of the change, so the correction follows the physical state of each detector unit.

The outputs are the corrected class counts, written back to the count archive and supplied to traffic management in place of the received counts, and a maintenance record holding, for each detector unit, its length offset, fault class, the part of the offset due to loop spacing, its bias at every boundary, and any dated change.

7. Detailed Description of the Invention

The system (900) and the roadside hardware it serves are shown in Figure 2, the sequence of operations in Figure 3, and the reference numerals in Table 1. The operating values below are those of the embodiment and of every measurement in Section 8.

Ref	Element	Ref	Element
100	roadside detector hardware	340	refit scheduler
110	detector site, loop pair in each lane	350	offset estimator
111	detector card	360	fault classifier
112	neighbouring detector sites	400	reclassification unit
120	slip road	405	interval total kept
130	communication link	810	corrected dataset

Ref	Element	Ref	Element
200	count archive	820	traffic-management process
300	pair-admission unit	830	maintenance record
310	share-difference unit	840	sensitivity correction of detector unit
320	chain solver	900	data-processing system
330	step-event detector		

Table 1: Reference Numerals Used in the Description and the Figures

Detector sites (110). Each site has a pair of inductive loops in every lane and a detector card (111) in a roadside cabinet that computes each vehicle's length and counts it into one of four classes separated by boundaries at 520, 660 and 1160 cm. Every fifteen minutes the card reports the four class counts, the total count and the mean speed.

Communication link (130) and count archive (200). The records travel over the link to the archive, which also holds the site register: the order of the sites along the carriageway and the positions of the slip roads (120). In the embodiment both are read from the site identifiers, whose marker number is the chainage and whose marker letter distinguishes mainline from slip-road sites.

Pair-admission unit (300). Consecutive mainline sites no more than 3 km apart with no slip-road site between them are candidates. An interval is used if both sites reported it, both report a mean speed above 50 mph, and the upstream site counted at least 100 vehicles; the speed condition keeps to free flow, where dual-loop length measurement is valid (entry 5). A candidate is admitted for a fitting period, one calendar month in the embodiment, if it has at least 500 such intervals and its two totals agree within 3 %. Figure 4(c) shows why: the largest difference among the M25 candidates, 38.9 points, belongs to a pair whose totals differ by 27 %, where the two sites do not see the same vehicles.

Share-difference unit (310). For each admitted pair and each boundary, the share below the boundary at each site is the sum of its counts below the boundary over the used intervals divided by the sum of all its counts, and g is the upstream share less the downstream share, over the whole period and day by day over days with at least eight used intervals. Figure 4(a) gives the spread of g at 520 cm across the candidate pairs of March, and Figure 4(b) its persistence: r = 0.987 between March and June on the M25 and 0.748 on the M1.

Chain solver (320). Consecutive admitted pairs sharing a site form a chain; a pair failing admission breaks it. For each chain of at least two pairs and each boundary, relative biases are accumulated from the first site, b_(i+1) = b_i - g_i, and the chain median is subtracted. The same solution is formed for each day on which every pair of the chain has data, and each site's standard error is the standard deviation of its daily values divided by the square root of their number: 0.08 points at 520 cm (median, M25) against a median bias magnitude of 2.5 points (Figure 4(d)).

The median reference has a measured limit. With a known fault injected into one real detector's record, the injected site shows the largest change on its chain in 80 % to 97 % of injections by chain length (Figure 6, Table 5); the median error in the fitted size is close to zero, but the 90th percentile is 27 % to 72 % of the fault, because a fault near the middle of a short chain moves the median itself. Agreement between neighbours does not depend on the chain's constant; absolute size on short chains does, and a verified site removes the limit.

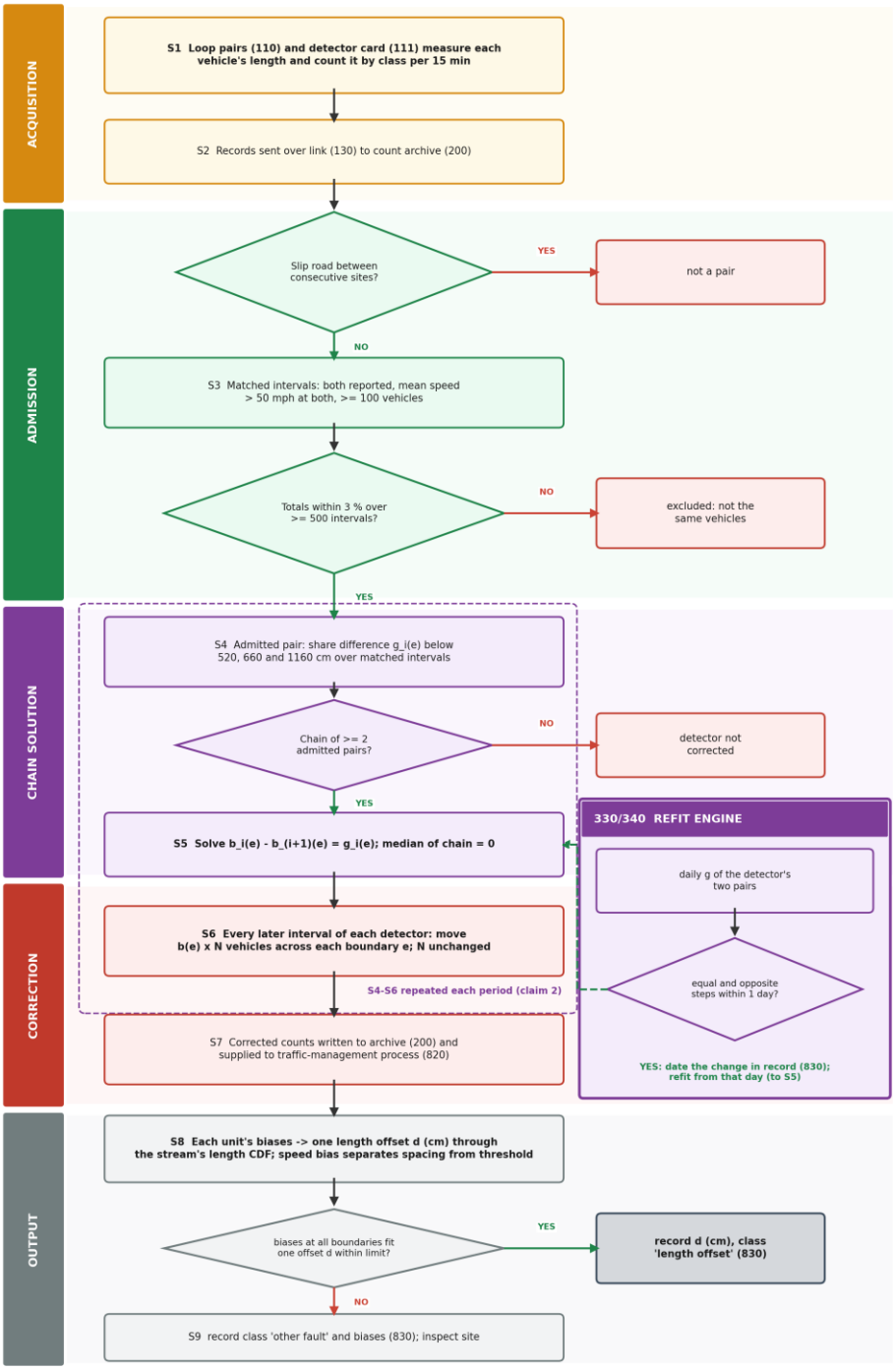


Figure 3. Sequence of operations from the loop pairs to the maintenance record. Acquisition (S1, S2), admission of pairs (S3 and two decisions), chain solution (S4, S5), correction (S6, S7) and output (S8, S9): each unit's biases become one length offset and a fault class; the refit engine (330, 340) dates changes and restarts S5, and S4 to S6 repeat each fitting period.

Offset estimator (350). The cumulative length distribution of the stream is interpolated through the median class shares of the fitted detectors, with a monotone piecewise-cubic (PCHIP) curve anchored at zero below 250 cm and at one at 2600 cm, above the longest permitted vehicle. Each unit's offset d solves F(520 - d) -

F(520) = b(520) by bisection. Over all fitted units with a bias at 520 cm above 2 points, the median |d| is 31 to 36 cm and the 90th percentile 72 to 100 cm across the eight corridor-months (Table 6, Figure 7(c)).

The single-offset structure is measured, not assumed. Taking all fitted units of a corridor-month as rows and the three boundaries as columns, one component explains 97.2 % to 98.8 % of the variance of the bias matrix in all eight corridor-months. Its ratio between the 660 cm and 520 cm boundaries is 0.17-0.18 on the M25 and 0.15-0.19 on the M1, against 0.18-0.21 and 0.20-0.22 predicted from the length density of the interpolated distribution (Figure 7(a), (b)); the prediction is higher than the fit by 0.01 to 0.05 in every corridor-month, a consistent gap that the coarse interpolation between class boundaries would produce. The prediction at 1160 cm is looser, since the interpolation has no class boundary between 1160 cm and its upper anchor.

Fault classifier (360). Each unit's bias vector is compared with the single-offset prediction, the unit's weight times the corridor's boundary loadings; where the largest departure exceeds 0.3 of the bias at 520 cm, the fault is classified as not a length offset. In March this applies to 23 of 130 M1 units with a bias above 2 points and to 2 of 61 on the M25. The class is a property of the unit: of the M1 units with a bias above 2 points in both March and June, 15 were classified so in March and 14 of them again in June (Figure 7(d)).

Speed attribution. Each unit's speed bias is solved along the chain from the log ratio of the mean speeds of each pair, with the median gauge. A spacing error large enough to shift speed by a factor k shifts length at 520 cm by about (k - 1) x 720 cm with 2 m loops. The median offset such speed biases could explain is 4 to 8 cm, against median offsets of 31 to 36 cm, and 44 % to 64 % of units with a bias above 2 points show a speed bias under 1 % (Figure 7(c)); most offsets are therefore detection-threshold errors, corrected by the detector's sensitivity or length constant rather than its loop spacing.

Reclassification unit (400). For every interval of a site with a fitted bias, the count below each boundary is set to round((s - b) n), where s is the interval's share below the boundary, b the site's bias there and n the interval total, bounded to [0, n] and kept non-decreasing across boundaries; the class counts are the successive differences. The total count (405) and the mean speed are not altered. The operation takes 6.6 microseconds per fifteen-minute record on one desktop processor (Table 2), against a reporting interval of 900 s.

The correction restores agreement between sites. On the longest May chain of the M25, 16 detector sites carrying the same traffic, the share below 520 cm ranges over 27.1 points as recorded and 1.0 points after correction with April biases (Figure 5(a)). Biases fitted on March remove 68 % to 92 % of the disagreement over the pairs touching a corrected detector in April, May and June, and 57 % to 79 % over all admitted pairs, including pairs whose detectors lie on chains too short to be corrected (Table 3); the other boundaries follow (Figure 5(d)).

Two alternatives were measured against the per-boundary correction. The proportional form leaves more disagreement in every month on both corridors (in June 2.65 against 2.91 points on the M1, Figure 5(c)). The single-offset form matches it at 520 cm but not at the smaller boundaries: in April on the M25 it leaves 0.67 points at 660 cm against 0.22 (Table 6). The offset is therefore the unit's physical correction and diagnosis, and the per-boundary biases correct the record.

Corrected dataset (810) and supply (820). The corrected counts are written to the archive in place of the received class counts. In live operation the same reclassification is applied to each record from a detector card as it arrives, with the biases of the latest fit, and the corrected record is supplied to the traffic-management process; the results of Section 8 are this computation applied one to three months after the fit.

Maintenance record (830). Each unit's offset, fault class, spacing part, biases, standard errors and fitting period are written to the record. A unit may be marked for attendance where its bias at 520 cm exceeds three standard errors and a tolerance set by the operator; at 1 point the mark falls on 74 of 113 fitted M25 units in March and at 5 points on 37 (Figure 9(c)). A local rule, marking a unit only when both its pairs disagree beyond the tolerance, disturbs other marks less often (0.31 against 0.52 per injected fault on the M25 at 2 points) but newly marks only 103 of 304 previously unmarked injected sites against 183 of 196 (Figure 9(d)).

Step-event detector (330). For each site belonging to two candidate pairs, the daily difference series of each pair at 520 cm is searched for its largest step, the split maximising the difference of the means on either side scaled by the pooled standard deviation, with at least 5 days on each side. An event is attributed to the site when the two steps fall within 1 day of each other, each exceeds 3 points, they are of opposite sign and their sum is under 30 % of the larger. Of 235 sites tested, 8 show the signature and the same rule applied to 4700 mismatched pairings finds 0 (Figure 8, Table 7). At M1 site 332 the pairs step by -30.4 and +33.8 points on 4 June 2025. The steps appear at 660 cm as well, and in 7 of the 7 events with both pairs measured, the two pairs agree on the ratio of the 660 cm step to the 520 cm step within 0.1, as one change of one unit should.

Refit scheduler (340). All biases and offsets are refitted each period and applied to the next, and a site with a dated event is refitted from the day of the event. In June, a fit on May leaves 1.43 points of disagreement on the M1 against 2.65 for a fit on March, and 0.46 against 0.92 on the M25 (Table 4, Figure 5(b)).

Sensitivity correction (840) and platform. The offset of a unit is the length the detector card should subtract, or the loop constant it should change, and the reclassification unit applies the biases to every record of the unit until it is retuned; the retune then shows as an equal and opposite step, and the refit takes the unit's new state from that day. The system (900) is a server or desktop processor attached to the archive: a month's fit takes 8.2 s for April's 125 admitted M25 pairs and 16.4 s for the M1's 201.

7.1 Alternative embodiments

Reference site. Where one site of a chain has had its length measurement verified with probe vehicles, its bias may be set to zero in place of the median condition, which gives the chain, and every unit's offset, an absolute calibration.

Other corrections and length distributions. Vehicles may be moved in proportion to the share below the boundary, or by the single offset alone; both were measured and are inferior on the data of Section 8. The length distribution may be interpolated by other monotone curves or taken from a classifier with more boundaries.

Other detector hardware, class schemes, fitting periods and networks: radar, video, piezoelectric and axle-based classifiers; any number of length classes; per-lane records; a week or a rolling window; any road on which consecutive detector sites observe the same traffic stream, including classified-count stations on national highways.

8. Validation - Study on Recorded Detector Data

The method was evaluated on field records from motorway inductive-loop detector sites: fifteen-minute class counts, totals and mean speeds from the WebTRIS service of National Highways, England, for carriageway A of the M25 (reference) and of the M1 (replication), March to June 2025. Biases are fitted on one month and scored on later months the fit did not see. The score is agreement between adjacent sites that measure the same vehicles, reported on two populations: the pairs touching a corrected detector, and all admitted pairs. Since totals are kept, the correction cannot change which pairs are admitted. No independent record of

vehicle lengths exists for these sites, so the evidence for the correction is agreement, and the evidence for the offsets is their internal physical consistency, not a measured length.

Performance metric	M25	M1
Candidate pairs in March (no slip road between, enough data)	112	256
Difference below 520 cm between paired sites, median (March), points	5.0	5.8
Admitted pairs, April / May / June	125 / 129 / 117	201 / 205 / 210
All admitted pairs, mean difference below 520 cm, recorded -> corrected (Apr-Jun), points	6.44-6.80 -> 1.83-2.15	7.90-8.69 -> 1.79-3.37
Pairs touching a corrected detector, the same, points	6.97-7.12 -> 0.58-0.92	8.18-8.94 -> 0.92-2.65
Variance of unit biases explained by one offset per unit, Mar-Jun	98.2-98.8 %	97.2-98.6 %
Length offset of units with bias > 2 points, median / 90th pct (March), cm	36 / 100	35 / 91
Injected faults: injected site shows largest change, range over fault sizes	84-96 %	82-96 %
Dated changes / sites tested; the same in control pairings	2 / 90; 0 / 1800	6 / 145; 0 / 2900
Fit of one corridor-month (April admitted pairs), s	8.2 (125)	16.4 (201)
Reclassification time per 15-min record, microseconds	6.57	6.37

Table 2: System Performance Metrics on Recorded Motorway Detector Data

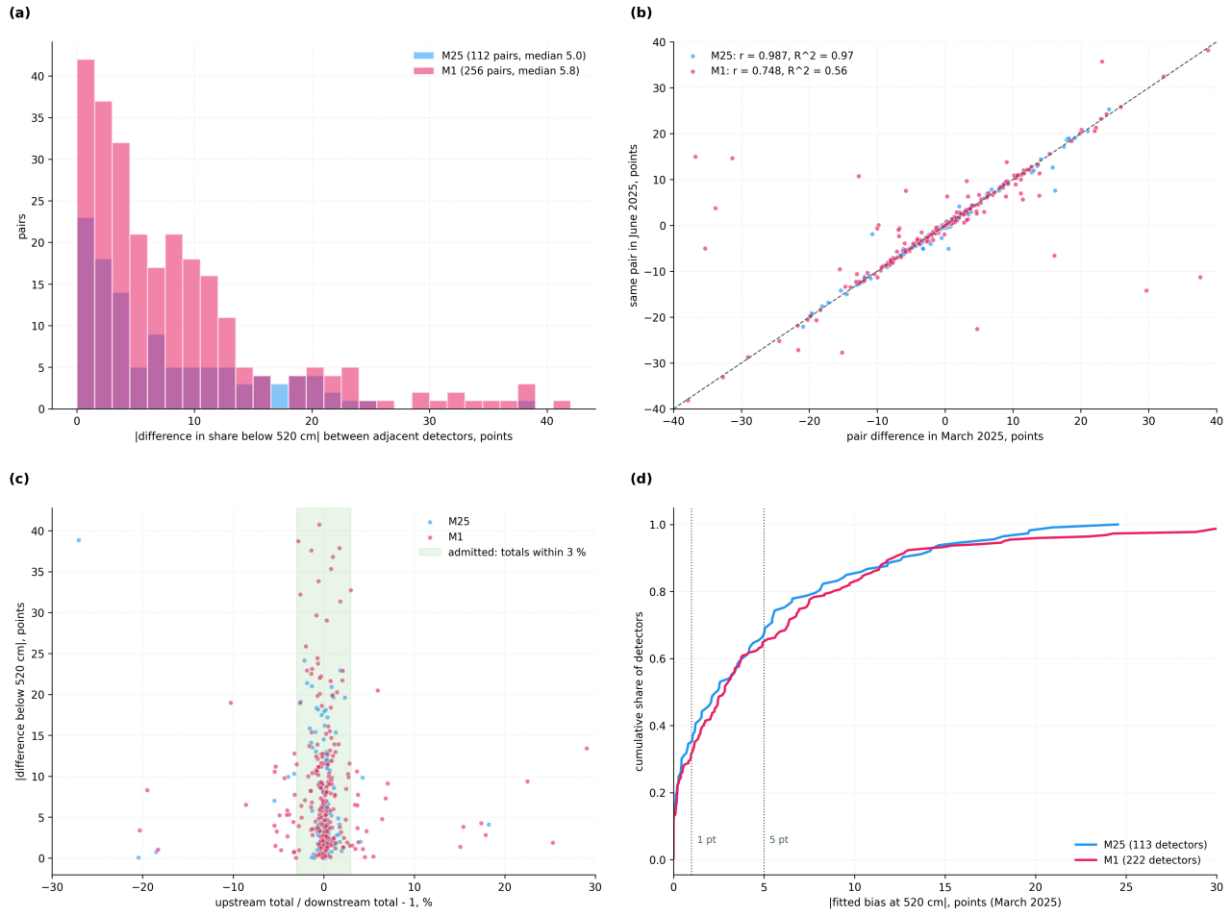


Figure 4. Disagreement in length classification between adjacent detector sites that count the same vehicles. (a) Differences in share below 520 cm across the candidate pairs of March; (b) the same pairs in March and June, showing the difference persists as a property of the sites; (c) difference against the ratio of totals, with the admission band; (d) fitted biases of all units on chains of three or more.

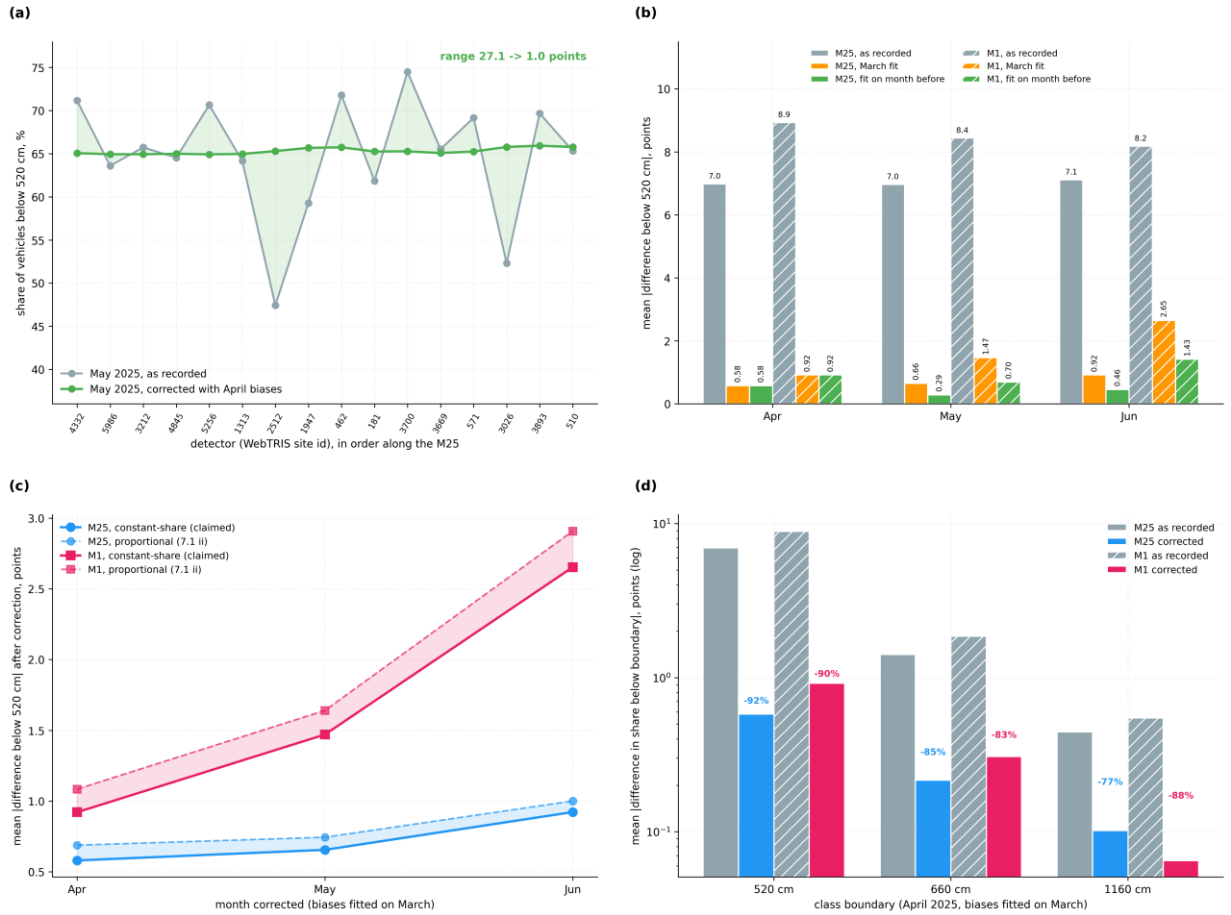


Figure 5. Correction of months the fit did not see. (a) One chain of the M25 in May, as recorded and corrected with April biases; (b) mean difference over pairs touching a corrected detector, as recorded, with a March fit and with a fit on the month before; (c) the claimed per-boundary correction against the proportional alternative; (d) all three boundaries in April.

Corridor	Month	Touched pairs	Recorded	Claimed correction	Proportional (7.1)	All pairs	All pairs: recorded -> corrected
M25	April	95	6.99	0.58	0.69	125	6.80 -> 1.93
M25	May	94	6.97	0.66	0.74	129	6.44 -> 1.83
M25	June	83	7.12	0.92	1.00	117	6.54 -> 2.15
M1	April	173	8.94	0.92	1.08	201	8.69 -> 1.79
M1	May	169	8.44	1.47	1.64	205	8.07 -> 2.33
M1	June	172	8.18	2.65	2.91	210	7.90 -> 3.37

Table 3: Mean Difference Below 520 cm Between Adjacent Sites, Points, Biases Fitted on March

Corridor	Month	Recorded	Fit on March	Worse	Recorded	Fit on month before	Worse
M25	April	6.99	0.58	3	6.99	0.58	3
M25	May	6.97	0.66	3	6.82	0.29	1
M25	June	7.12	0.92	2	6.44	0.46	0
M1	April	8.94	0.92	6	8.94	0.92	6
M1	May	8.44	1.47	9	8.36	0.70	4
M1	June	8.18	2.65	14	8.16	1.43	8

Table 4: Fixed Fit Against Refit on the Month Before, Pairs Touching a Corrected Detector, Points

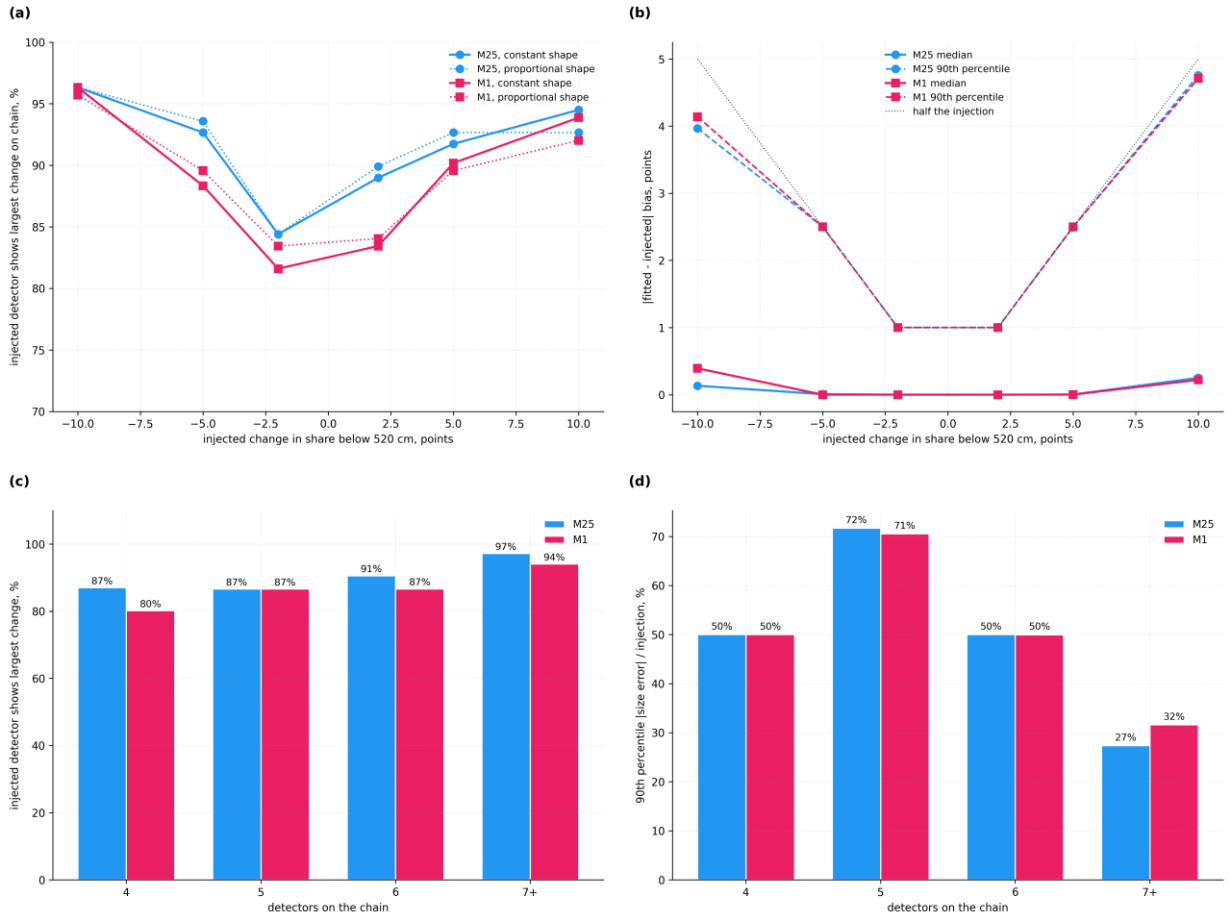


Figure 6. Known faults injected into the records of real detector sites. (a) Share of injections in which the injected site shows the largest change on its chain; (b) error in the fitted size; (c) and (d) the same by chain length, where the median reference limits absolute size on short chains.

Corridor	Sites on chain	Injections	Largest change at injected site	Size error / injection, median	90th percentile
M25	4	240	87 %	13 %	50 %
M25	5	300	87 %	0 %	72 %
M25	6	288	91 %	0 %	50 %
M25	7+	480	97 %	0 %	27 %
M1	4	384	80 %	18 %	50 %
M1	5	120	87 %	0 %	71 %
M1	6	504	87 %	0 %	50 %
M1	7+	948	94 %	0 %	32 %

Table 5: Injected Faults by Chain Length, Both Shapes and All Sizes Pooled

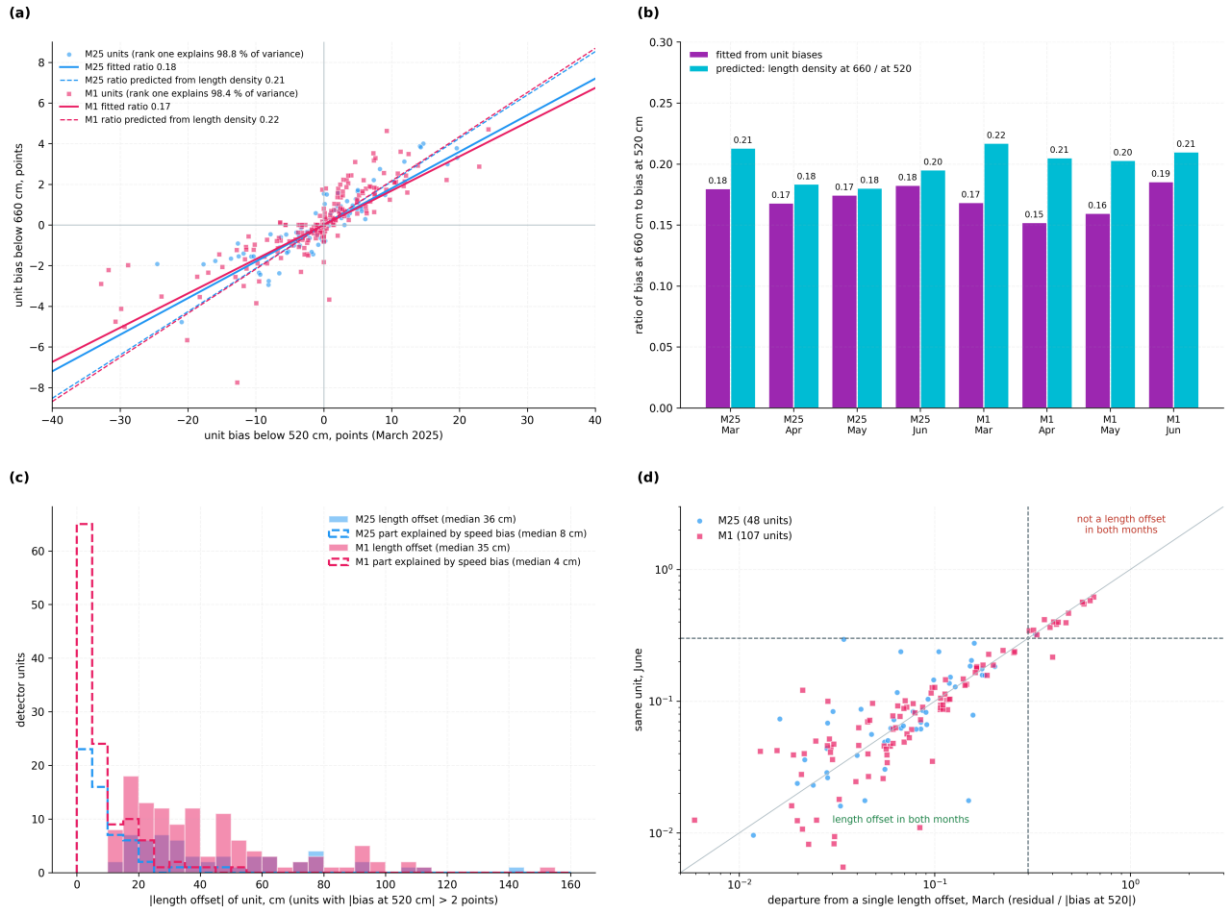


Figure 7. One length offset per detector unit, recovered from class counts. (a) Bias at 660 cm against bias at 520 cm for every fitted unit, with the fitted ratio and the ratio predicted from the stream's length density; (b) the two ratios in every corridor-month; (c) offsets in cm and the part a speed error could explain; (d) departure from a single offset in March and June, the fault class persisting.

Corridor	Month	Units	Variance, one offset	Ratio 660/520 fitted	Ratio predicted	|Offset| median / p90, cm	Not an offset / units > 2 pt	Offset from speed, median cm
M25	March	113	98.8 %	0.180	0.213	36 / 100	2 / 61	7.6
M25	April	130	98.6 %	0.168	0.184	32 / 86	1 / 67	7.4
M25	May	136	98.6 %	0.175	0.180	31 / 87	1 / 63	7.0
M25	June	122	98.2 %	0.183	0.195	31 / 84	3 / 61	7.8
M1	March	222	98.4 %	0.169	0.217	35 / 91	23 / 130	4.4
M1	April	220	98.6 %	0.152	0.205	36 / 84	15 / 126	5.0
M1	May	219	97.9 %	0.160	0.203	34 / 73	19 / 125	5.2
M1	June	225	97.2 %	0.186	0.210	35 / 72	21 / 124	4.6

Table 6: The Single Length-Offset Model in Every Corridor-Month

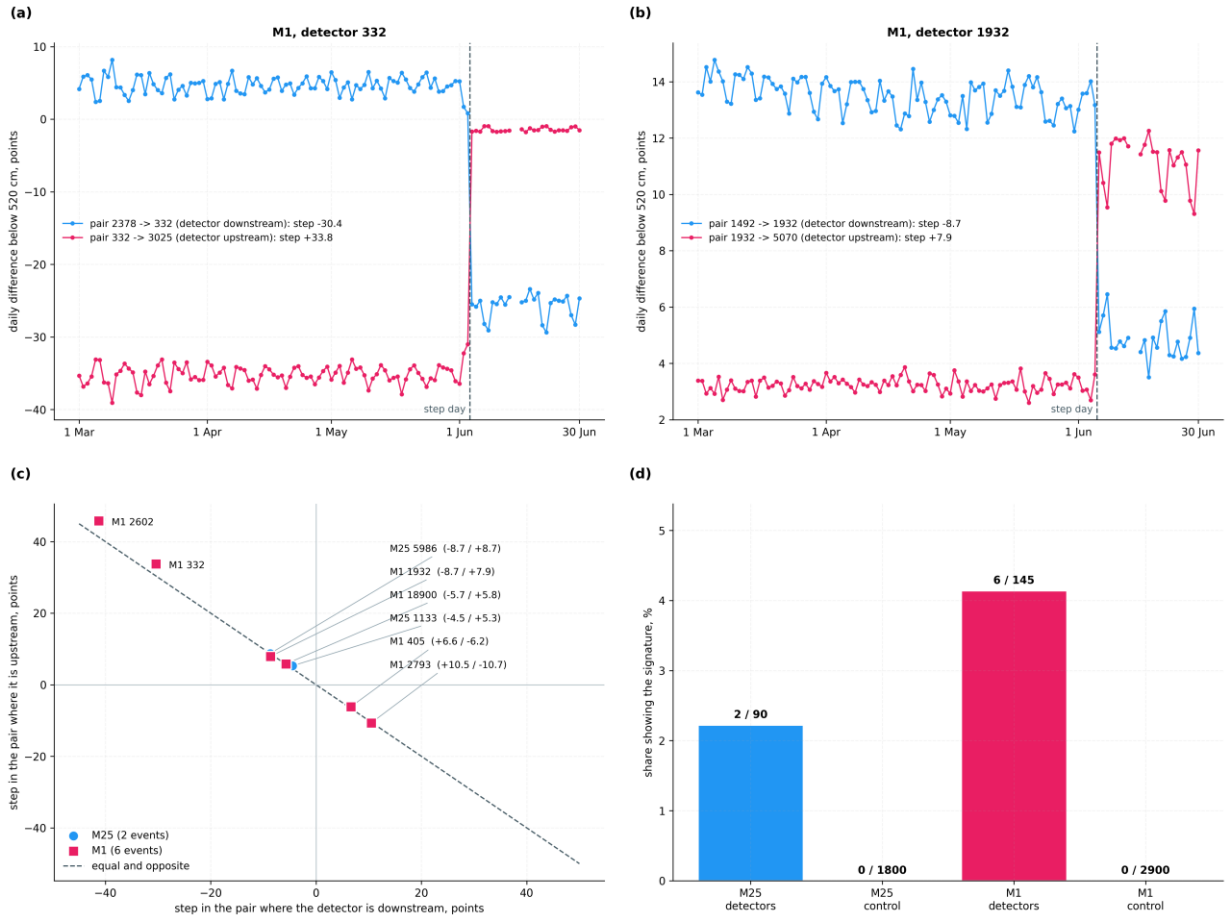


Figure 8. Dated changes in length measurement attributed to one detector unit. (a), (b) Daily differences of the two pairs of M1 sites 332 and 1932, stepping on the same day by equal and opposite amounts; (c) the two steps of every event against the equal-and-opposite line; (d) the signature in tested sites and in mismatched control pairings.

Corridor	Site	Day of step	Pair where downstream, pts	Pair where upstream, pts	Step ratio 660/520, both pairs	Data within 5 days
M25	1133	16 June 2025	-4.5	+5.3	0.26 / 0.22	site gave no usable data on 2 days
M25	5986	1 April 2025	-8.7	+8.7	0.18 / 0.16	site gave no usable data on 8 days
M1	18900	26 March 2025	-5.7	+5.8	0.05 / 0.06	both pairs report every day
M1	1932	6 June 2025	-8.7	+7.9	0.23 / 0.30	both pairs report every day
M1	2602	14 May 2025	-41.3	+45.8	- / 0.14	one pair missing before the step
M1	2793	3 April 2025	+10.5	-10.7	0.09 / 0.09	site gave no usable data on 6 days
M1	332	4 June 2025	-30.4	+33.8	0.11 / 0.13	both pairs report every day
M1	405	5 April 2025	+6.6	-6.2	0.04 / -0.05	site gave no usable data on 2 days

Table 7: Changes in Length Measurement Attributed to One Site and One Day

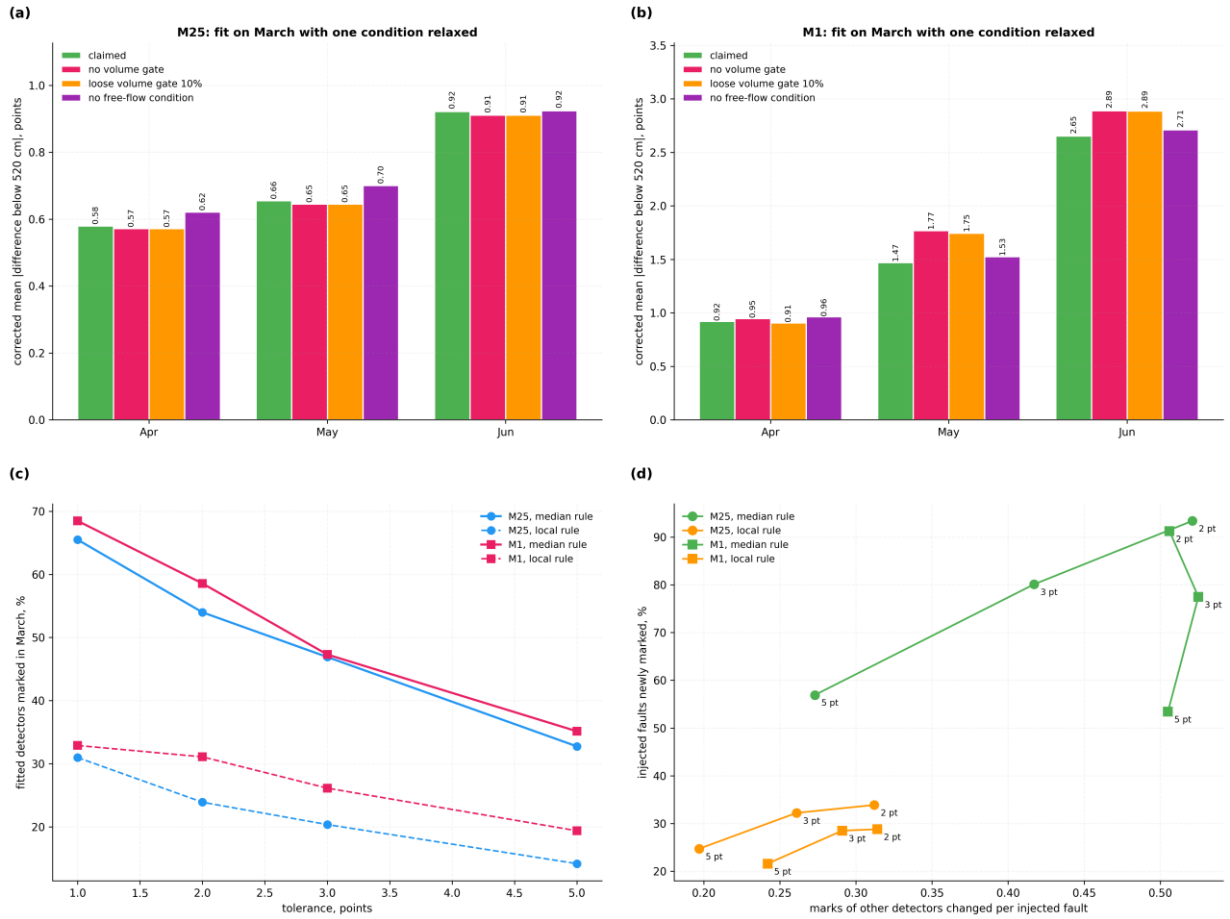


Figure 9. Admission conditions and the attendance mark. (a), (b) Corrected disagreement when the fit is made with the balance test or the free-flow condition relaxed; (c) share of fitted units marked for attendance at each tolerance under the median and local rules; (d) injected faults newly marked against marks of other units disturbed.

Capability	Entry 1	Entry 2	Entry 3	Entry 7	Present invention
Input already in the archive	Aggregated AEVL	No: per-vehicle event data	No: probe runs	Sensor readings	Yes: class counts
Reference for calibration	Adjacent stations	External length distribution	Probe vehicles of known length	Neighbouring sensors	Neighbouring sites, same vehicles
Localises the faulty site	Yes	Per station	Site chosen for visit	-	Yes
Signed error at each class boundary	No	No	No	No	Yes
Physical length offset of the unit, cm	No	Sensitivity level from event data	From probe runs at the site	No	Yes: from class counts
Tells an offset from another fault	No	No	No	No	Yes
Corrects the recorded class counts, totals kept	No	No: retunes the loop	No	Readings, not counts	Yes
Needs attendance at the site	For recalibration	Yes: laptop at cabinet	Yes	No	No
Dates a change in a site's measurement	No	Compares before and after a known tune-up	No	No	Yes

Table 8: Capabilities Against the Closest Prior Art (Numbering of Table A)

The limits of these results are stated here. Agreement between corrected neighbours is established; whether a chain's median site is correct is not, and a verified reference site addresses it. The offsets in centimetres rest

on the interpolated length distribution and are physically consistent across boundaries, months and corridors, but have not been compared with measured lengths. Only sites on chains of three or more are corrected, which is why the reduction over all admitted pairs is smaller than over the pairs touched. Absolute size on short chains carries the error of Table 5. Biases change, so the fit must be refreshed. The data cover two corridors, one carriageway each, four months and one national detector network.

9. Contemplated Prototype and Planned Validation

The method is implemented and runs end to end on archived field records (Section 8). Not yet done: operation on a live detector feed, and comparison with independently measured vehicle lengths.

#	Planned measurement	What it would establish	Claims
1	Class shares and offsets at a site co-located with a weigh-in-motion or camera classification station, or after a probe-vehicle calibration	absolute accuracy of the corrected record and of the offsets in cm	1, 5, 6
2	Maintenance logs for the sites of Table 7 and the units classed as not an offset	that dated events and fault classes match work done at the site	3, 5
3	Live operation, correction per reporting interval at the archive	latency and continuity in service	4, 7
4	A second national network, for example classified counts on an Indian national highway	transfer to other detector hardware and fleet mixes	1, 7

Table 9: Planned Measurements

10. Aspects of the Invention Requiring Protection (Claims)

1. Admission of consecutive roadside detector-site pairs without an intervening slip road, confirmed by agreement of their total counts.

2. Signed length-class bias per detector site and per class boundary, solved along a chain of admitted pairs.

3. Length offset in centimetres of each detector unit, recovered from class counts through the stream's own length distribution, with a fault class.

4. Total-preserving reclassification of each site's archived and incoming length-class counts by its fitted boundary biases.

5. Data-processing system coupling inductive-loop detector sites, count archive and maintenance record of offsets and fault classes.

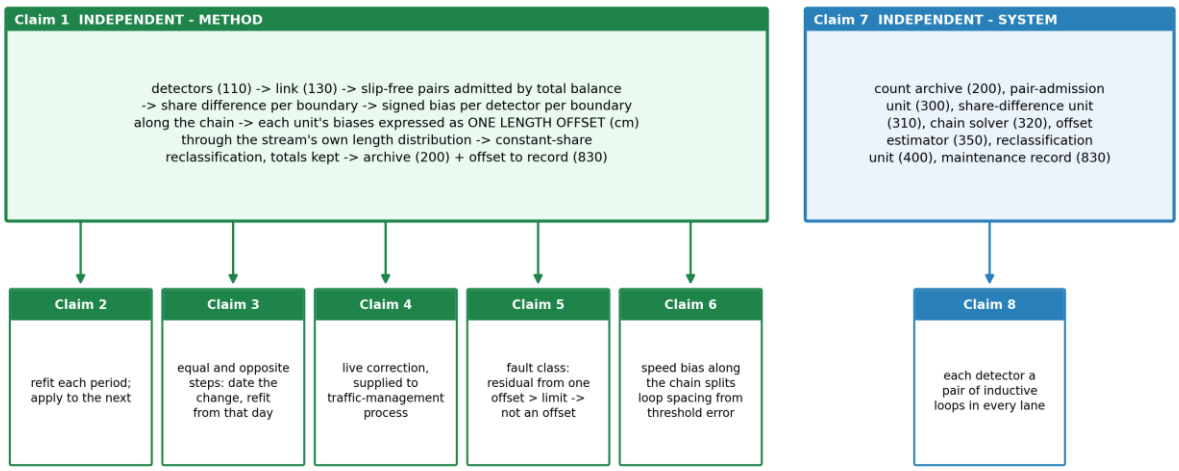


Figure 10. Claim tree. One independent method claim with five dependents, and one independent system claim with one dependent.

A. Method claims

Claim 1 (independent). A method of correcting the length classification of a plurality of roadside vehicle detectors (110) disposed in sequence along a carriageway, carried out by a data-processing system (900) connected to the detectors by a communication link (130), each detector measuring the length of each vehicle passing it and reporting, for each of a succession of measurement intervals, the number of vehicles in each of a plurality of length classes separated by length-class boundaries, the method comprising: (a) receiving, from the detectors over the communication link (130), their per-interval length-class counts, together with information identifying the order of the detectors along the carriageway and the consecutive detectors between which a slip road (120) joins or leaves the carriageway; (b) selecting each pair of consecutive detectors between which no slip road joins or leaves the carriageway, and admitting the pair for a fitting period only if, over the intervals of that period in which both detectors reported, their total counts agree within a balance tolerance, whereby the two detectors of an admitted pair measure the same vehicles, whose lengths do not change between them; (c) for each admitted pair and each length-class boundary, computing over those intervals the difference between the shares of vehicles that the two detectors report below the boundary; (d) for each chain of at least two admitted pairs in which consecutive pairs share a detector, determining for every detector of the chain, separately for each boundary, a signed share bias such that for each pair of the chain the bias of its upstream detector less the bias of its downstream detector equals the share difference of that pair, the constant common to the chain being fixed by setting to zero either the median of the biases over the chain or the bias of a detector on the chain whose length measurement has been verified; (e) estimating a cumulative length distribution of the vehicles passing the chain by interpolating, through the length-class boundaries, the shares of vehicles that the detectors of the chain report below those boundaries, and determining for each detector a length offset, being the distance by which the detector over-measures each vehicle length, such that the change in the estimated cumulative distribution at a first boundary when that boundary is reduced by the offset equals the detector's share bias at that boundary; (f) generating a corrected length-class count dataset (810) for measurement intervals after the fitting period by reclassifying, in each interval of each detector of the chain, a number of vehicles across each boundary equal to that detector's share bias at the boundary multiplied by the total count of the interval, the total count of every interval being left unchanged; and (g) storing the corrected dataset in a count archive (200) in place of the received length-class counts, and writing to a maintenance record (830), for each detector of the chain, its identity and its length offset as a sensitivity correction for that detector unit.

Claim 2. The method of claim 1, wherein steps (b) to (f) are repeated for successive fitting periods, the biases and offsets determined over each fitting period being applied to the measurement intervals of the period that follows it.

Claim 3. The method of claim 1, further comprising, for each detector belonging to two admitted pairs: forming for each of the two pairs the daily series of its share difference below a boundary; locating in each series the split of largest difference of means relative to the pooled standard deviation; and, where the two splits fall within one day of each other, each step exceeds a minimum size, they are of opposite sign and their sum is less than a set fraction of the larger, writing to the maintenance record a change in the length measurement of that detector dated to the day of the steps, and determining that detector's biases and offset from measurement intervals after that day.

Claim 4. The method of claim 1, wherein step (f) is applied to the length-class counts received from each detector after the fitting period as they are received, and the corrected counts are supplied to a downstream traffic-management process (820) in place of the received counts.

Claim 5. The method of claim 1, further comprising determining, for the detectors of one or more chains, a loading per boundary common to the detectors as the principal direction of their bias vectors across the boundaries, and classifying each detector in the maintenance record as having a length-offset fault where the largest departure of its bias vector from its projection on that direction is within a set fraction of its bias at the first boundary, and as having a fault of another kind otherwise.

Claim 6. The method of claim 1, further comprising determining for each detector of the chain a speed bias, from the ratios of the mean speeds reported by the two detectors of each admitted pair solved along the chain as in step (d), and writing to the maintenance record the part of the detector's length offset attributable to a loop-spacing error, computed from the speed bias, and the remainder as a detection-threshold error.

B. System claims

Claim 7 (independent). A data-processing system (900) for correcting the length classification of roadside vehicle detectors (110) disposed in sequence along a carriageway, comprising: a count archive (200) receiving over a communication link (130) the per-interval length-class counts of the detectors; a pair-admission unit (300) configured to carry out step (b) of claim 1; a share-difference unit (310) configured to carry out step (c) of claim 1; a chain solver (320) configured to carry out step (d) of claim 1; an offset estimator (350) configured to carry out step (e) of claim 1; a reclassification unit (400) configured to carry out step (f) of claim 1 and to write the corrected dataset to the count archive; and a maintenance record (830) holding the length offset of each detector unit.

Claim 8. The system of claim 7, further comprising the detectors, each detector comprising a pair of inductive loops in each lane of the carriageway and computing the length of each vehicle from the times at which the vehicle occupies the two loops.

Evidence for each claim: Claim 1, Tables 2, 3, 5 and 6, Figures 4 to 7; Claim 2, Table 4 and Figure 5(b); Claim 3, Table 7 and Figure 8; Claim 4, Table 3 (the correction of later months is this computation) and the latency of Table 2; Claim 5, Table 6 and Figure 7(d); Claim 6, Table 6 and Figure 7(c); Claims 7 and 8, Section 7.

11. Technology Readiness Level (TRL)

The method is implemented in full and validated on archived field records from two motorway corridors at full corridor scale, each claimed element measured out of sample, against injected faults, against a control or for physical consistency (Section 8). It has not run on a live feed or been compared with measured vehicle lengths. Readiness is assessed at TRL 4, technology validated in the laboratory; measurements 1 and 3 of Table 9 would support TRL 5.

Research			Development			Deployment		
TRL 1	**TRL 2**	**TRL 3**	**TRL 4**	**TRL 5**	**TRL 6**	**TRL 7**	**TRL 8**	**TRL 9**
Basic principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in lab	Validated in relevant environment	Demonstrated in relevant environment	Prototype in operational environment	System complete and qualified	Proven in operational environment
✓	✓	✓	✓					

Current TRL: Level 4 — Technology Validated in Laboratory.

12. Note on Filing

Five matters are flagged for the IPR & TT Cell.

First: technical character. Claim 1 receives the output of roadside measuring hardware, relies on a physical invariant of the vehicles that hardware measures, determines a physical length offset of each detector unit in

centimetres, and changes both the record the detectors produce and the maintenance record of the units. No claim rests on a planning or control effect.

Second: inventive step. The attack to be expected combines entry 1 (comparison and flagging between adjacent stations) with entries 7 and 8 (calibration against neighbours) and entry 2 (retuning from the length distribution). The answer relied on is element (d) of Section 4.2: recovering each unit's physical offset from class counts through the stream's own length distribution, which rests on a measured single-offset structure whose boundary ratio is predicted from the counts themselves, and which yields a fault class and a spacing-or-threshold attribution that no reference provides. Lu, Yang and Chang (2014), cited by entry 1, is closed access and was not read in full; it should be obtained and checked before filing.

Third: relation to the inventor's earlier disclosure on the conservation of total vehicle counts. Here total counts only admit a pair, and the correction keeps them; no claim relies on a total-count residual. The earlier disclosure is not prior art to this one until it is published, and this one should be filed before then.

Fourth: claim breadth and limits. Claim 1 recites the per-boundary correction, which was measured best, and the offset, whose centimetre value depends on the interpolated distribution and has not been compared with measured lengths; the chain's constant by a verified site is disclosed as the route to absolute accuracy but was not measured. Claim 3 rests on detection with a clean control but few events.

Fifth: inventors. Inventor details are given in the accompanying IDF-A.